

Hemanth Kumar Nalla

SDET | QA Automation Engineer

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PROFESSIONAL SUMMARY

QA Automation Engineer with 3+ years of experience across enterprise tax software on Thomson Reuters ONESOURCE Corporate Tax. Delivered 150 work items across 2025 and H1 2026 with zero production escapes, including validation of 17 critical fixes for key enterprise customers. Strong background in Playwright, TypeScript, Python, API testing, and regression strategy, with a track record of building AI-assisted QA workflows and reusable automation frameworks that speed up test-case generation, impact analysis, and coverage discovery.

KEY ACHIEVEMENTS

- **Automation scale-up:** Built a Playwright framework spanning 39 tests, 26 specs, and 11 page objects across SAT, QA, EMEA, and APAC environments.
- **AI-driven QA tooling:** Designed Claude- and MCP-based workflows that map Azure DevOps work items and GitHub PRs to impacted modules and auto-generate structured test cases, cutting the manual effort of impact analysis and test-case authoring.
- **Coverage intelligence:** Introduced a RAG-based semantic-search layer over existing test assets, accelerating coverage-gap analysis and reducing duplicate test coverage.
- **Team visibility:** Set up CI-driven API test reporting via GitHub Actions and GitHub Pages, giving the team a live, shareable results dashboard.

CORE SKILLS

- **AI for QA:** GitHub Copilot, Claude, MCP tooling, RAG-based test search, Streamlit, prompt engineering
- **Test Automation:** Playwright, Page Object Model, WebdriverIO, TestRigor
- **Testing:** Functional, Regression, API, End-to-End, Smoke Testing, Defect Verification
- **Languages:** Python, TypeScript, JavaScript, SQL
- **DevOps & Tools:** Azure DevOps, Git, GitHub, GitHub Actions, GitHub Pages, Linux
- **Practices:** Agile, Scrum, SDLC, STLC, root cause analysis, release support, cross-functional collaboration

PROFESSIONAL EXPERIENCE

Software Engineer – Quality Assurance

Oct 2024 – Present

Thomson Reuters International Pvt Ltd | Hyderabad, India

- Delivered 150 work items across 2025 and H1 2026 with zero production escapes, including validation of 17 critical fixes for key enterprise customers.
- Built AI-assisted QA workflows using Claude and MCP tooling to analyze Azure DevOps work items and linked GitHub PRs, mapping code changes to impacted modules and auto-generating structured test cases — reducing time spent on manual impact analysis.
- Built Playwright automation with AI-assisted generation and scaffolded execution, using a clean Page Object Model, reusable fixtures, and multi-environment support to scale regression coverage across SAT, QA, EMEA, and APAC.
- Created semantic-search workflows over existing test assets, accelerating coverage-gap analysis, deduplication, and triage.
- Set up CI-driven reporting for API test execution using GitHub Actions and GitHub Pages, delivering a shareable live results link for the team.
- Stabilized automation frameworks by improving 500-error handling, markup reliability, flaky smoke coverage, and regression execution consistency.
- Collaborated with development, DevOps, and product teams on release validation, defect reproduction, coverage planning, and production support.

Associate Software Engineer – QA (Contract)

Nov 2023 – Oct 2024

Ascent, engaged with Thomson Reuters | Hyderabad, India

- Executed manual, regression, and verification testing across sprint-based releases for the ONESOURCE Corporate Tax platform.
- Maintained and updated existing automation suites to keep regression coverage reliable across releases.
- Authored reusable prompt templates for recurring QA workflows and helped introduce GitHub Copilot into pull-request review and debugging support.
- Built the team's first TestRigor automated tests, contributing to faster code-level fixes during QA cycles.

Software Intern

Jan 2023 – Oct 2023

Thomson Reuters International Pvt Ltd | Hyderabad, India

- Built foundational QA skills through manual testing, test-case design, defect logging, and Agile delivery support.
- Performed smoke testing on new builds to confirm core system stability before deeper regression cycles.
- Worked closely with senior team members on test execution, issue validation, and release-readiness checks.

SELECTED PROJECTS

AI-Powered QA Suite for OCT

- Built an AI-assisted workflow (Claude + MCP) that reads Azure DevOps work items, pulls linked GitHub PRs, maps code changes to affected OCT modules, and auto-generates structured test cases — cutting manual effort in impact analysis.
- Added a RAG-based semantic-search layer over existing test cases to speed up coverage analysis, reduce duplication, and support root-cause investigation.
- *Stack: Python, Azure DevOps API, GitHub API, RAG embeddings, Streamlit, Claude MCP.*

AI-Generated Playwright Automation for OCT

- Built a Playwright automation framework with AI-assisted generation and scaffolded execution via the Playwright CLI — 39 tests across 26 specs and 11 page objects, supporting SAT, QA, EMEA, and APAC environments.
- Added reusable login fixtures plus screenshot and video reporting, cutting debugging time and improving execution transparency.
- *Stack: Playwright, TypeScript, Page Object Model, GitHub Actions, Claude.*

OCT Test Cases Open Arena

- Built a repeatable RAG pipeline that extracts Azure DevOps test cases from multiple suites and publishes curated datasets into Open Arena for semantic retrieval.
- Enabled AI-assisted reuse of historical test knowledge for faster test-case generation and discovery of similar coverage, reducing redundant authoring.
- *Stack: Python, Azure DevOps API, Open Arena, Claude Haiku, Excel, S3 pipeline.*

EDUCATION

Master of Computer Applications (MCA), Computer Science

Dec 2021 – Dec 2023

University College of Engineering, Osmania University | Hyderabad, India

B.Sc. – Mathematics, Computer Science

2018 – 2021

UPGCS, Osmania University | Hyderabad, India | CGPA: 8.23/10